



亞洲大學
ASIA UNIVERSITY

Midterm Project Report

Advanced Computer Programming

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Student ID : 114021142

Teacher : DINH-TRUNG VU

2026-04

Chapter 1 Introduction

1.1 GitHub

- 1) **Personal GitHub Account:** <https://github.com/adhamilton2700>
- 2) **Group Project Repository:** <https://github.com/wangyalal/Group1>

1.2 Overview

My program makes use of data class, the datetime library, and scrapy, with use of items and pipeline, for extracting information from the GitHub repository page. My personal GitHub currently solely has one repository, so in addition to that, <https://github.com/adhamilton2700?tab=repositories>, I also made use of the teacher's repository page, <https://github.com/trungvdhp?tab=repositories>, in order to better showcase the program's ability to handle data on a larger scale and also with larger variance in each information field. From each repository, the program pulls the respective username, URL, name, about information, languages used, amount of commits and also the date of the last update.

Chapter 2 Implementation

2.1 Class 1: GithubreposcraperSpider

The class which contains the name of the spider, the single specified rule, allowed domain("github.com"), and the two parse methods which dictate where to find and how to process the desired information. Imports data container from the folder's items file to be used for storage and output to the file.

2.1.1 Methods

The spider makes use of two methods:

- "parse" – this method identifies the username of the respective page, identifies every respective repository, getting their links and fetching the respective page so information can be extracted. It uses a for loop to look through each repository, first taking and storing the date of the last update to the repository. The method then calls the next method(rep_parse) and passes both the username and date of last update to it using callback keyword arguments
- "rep_parse" – the method responsible for extracting the information from the URL page of each repository found by the "parse" method. Starts by creating an instance of the data container(Repospideritem) to hold the extracted data, and uses CSS selectors to grab information for the repository name, about, languages, number of commits, and also grabs the URL and the variables passed from the "parse" method. Finally yields the item, sending it to the file.

2.2 Class 2: RepospiderItem

Container to be used for storing scraped data in a structured form

2.2.1 Fields

- “User” - the username of the repository’s owner
- “Url” - repository URL
- “Name” - name of the repository
- “About” - the description of the contents of the repository if any
- “Language” = a list of programming languages used in the repository
- “Commit_Count” = the number of commits to the repository
- “Last_Update” = the date of the most recent update to the repository

2.3 Class 3: RepospiderPipeline

Class for pipeline processing, imports “datetime” class. Processes data before allowing it to be stored and output to file.

2.3.1 Method/Function 1

- “Process_item method” – uses ItemAdapter to grab the string pulled by the program for the last update time, then uses “datetime.fromisoformat()” method to convert that string into a python readable time object. Then uses “strftime()” method to format that object into easily legible form; eg 03-May-2019 at 19:45:21. Ends by returning the item to be stored and output to file.

Chapter 3 Results

3.1 Result 1

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<User>adhamilton2700</User>

<Url>https://github.com/adhamilton2700/Resume</Url>

<Name>Resume</Name>

<About>AI written code in html for resume</About>

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<Name>caro-game-svn</Name>

<About>Automatically exported from code.google.com/p/caro-game-svn</About>

<Language>

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<value>HTML</value>

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<value>CSS</value>

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<Name>qlkho-dongluc</Name>

<About>Automatically exported from code.google.com/p/qlkho-dongluc</About>

<Language>

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<Name>netsharing</Name>

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<Name>cong-thong-tin-sv</Name>

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<value>C#</value>

<value>JavaScript</value>

<value>CSS</value>

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<Name>duongtinhduyen</Name>

<About>Automatically exported from code.google.com/p/duongtinhduyen</About>

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<Name>studen-portal</Name>

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<Name>master-ntou</Name>

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<About>joomla</About>

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<About>Deep Sky</About>

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<Name>mpsnet</Name>

<About>Mobile Palm Vein Recognition using Saturation channel and Deep Learning-MPSNet</About>

<Language>

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<Name>palmid</Name>

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<Name>ACP</Name>

<About>Repository for Advanced Computer Programming AU-1132</About>

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<Name>yolo-gradio-demo</Name>

<About>Demo for AI Web Apps: YOLO Object Detection on Gradio
Framework</About>

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<About>Cup bottle classifier</About>

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Chapter 4 Conclusions

The spider successfully scrapes information from a combined total of 25 repositories on both GitHub users' repositories, finding, storing and outputting values for each repository about the respective user, repository URL, repository name, the description(if any, or the repository name if none is found), a list containing the programming languages(if any) used, number of commits and date of last update, making use of the pipeline to format the date and time into a more clear and easily read form.